Scoring, substitution matrices and score aggregation for TCR antigen-specificity search

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vdjmatch technical appendix

Abstract

vdjmatch annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the seqtree search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (CASS...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by seqtree. The E-value theory used for significance is derived in the companion seqtree appendix; here we summarize it and give the paired-chain extension.

1. The search-and-score framework

Given a query CDR3 q and a labelled reference set D (VDJdb), vdjmatch retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the seqtree engine, which defines a ball $B_\theta(q) = \{x : \mathrm{edit}(q, x) \le \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (=/X/I/D), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§ 3).

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2. Substitution scoring and the VDJAM matrix

How BLOSUM is built, and why it does not transfer.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab}; (iv) score $s(a,b) = \frac{1}{\lambda}\log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a. The signal is "which residues substitute for one another within conserved, column-homologous positions," and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so "alignment columns" do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins `CASS` and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal `CASS` is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

What VDJAM changes (point by point).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only "aligned residue pair" obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a, a *position-specific* background $\mathrm{bg}(\cdot \mid L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM's percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are "expected" and contribute little), and prospectively by the control-calibrated significance layer (§ 3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

Observation unit and estimator.. We replace "alignment columns" with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c, let $\mathrm{bg}(\cdot \mid L, p)$ be the residue frequency at position p among all length-L chain-c CDR3s. With $f(a,b)$ the observed count of $\{a,b\}$ events and

$$e(a,b) = \sum_{\text{event slots } (L,p)} n_{L,p}\left[\mathrm{bg}(a \mid L,p)\,\mathrm{bg}(b \mid L,p) + \mathrm{bg}(b \mid L,p)\,\mathrm{bg}(a \mid L,p)\right], \qquad \mathrm{VDJAM}(a,b) = \log_2 \frac{f(a,b)+\alpha}{e(a,b)+\alpha}, \tag{1}$$

where $n_{L,p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and "expected" (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\mathrm{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

Region structure: NDN is learned, V/J are germline-determined.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank "substitutions" are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \Pr[\text{position } p \text{ is germline-retained}]$, determined by the gene's trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via `mirpy`'s `mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the `CASS` V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by $1 - \Pr[\text{germline}]$ and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

3. Score aggregation and significance

Per-hit to per-query.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

Control-calibrated E-value (primary).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query's VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M)\, n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

Paired $\alpha\beta$ E-value.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural

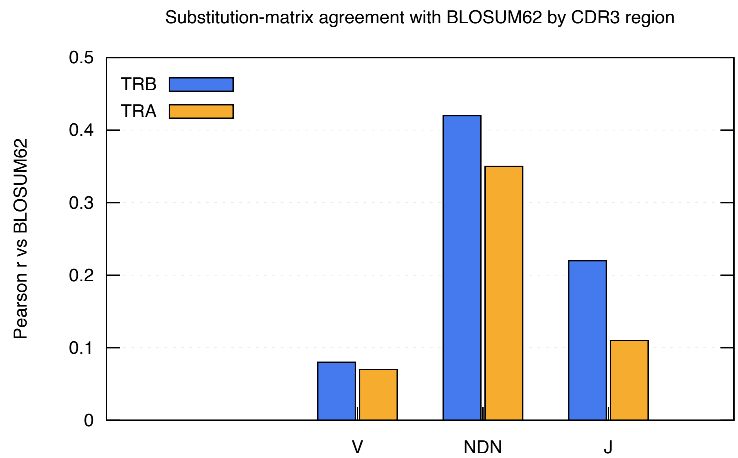

Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

and paired-sequencing data [4], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^{\alpha}\pi_0^{\beta}$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N\,\pi_0^{\alpha}\pi_0^{\beta}$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain's, true pairs are dramatically more significant: on the paired TCRvdb benchmark, $296/300$ pairs match a complex and $295/296$ (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9}. (When the two chains' hits are treated as independent tests, the equivalent statistic is Fisher's combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. Results: does VDJAM (and region weighting) help?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDJAM, and region-weighted NDN-VDJAM — ranking by score and reporting micro PR-AUC for "hit shares E^*" (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* ($+0.004$ at scope 2, $+0.007$ at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution*

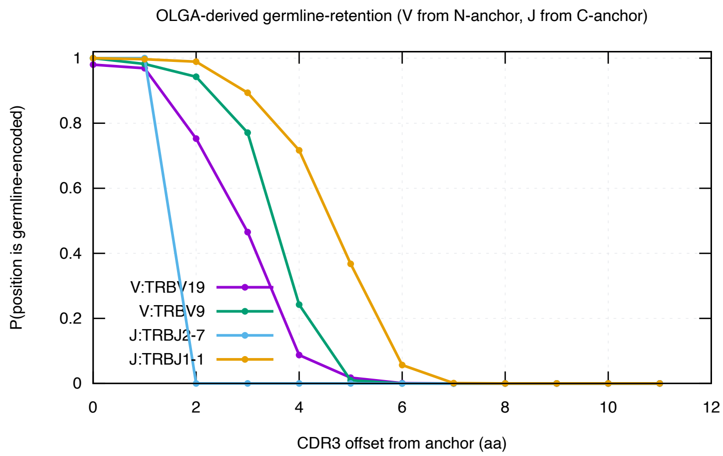

Figure 2: OLGA-derived germline-retention $w(p)$ (via `mirpy`): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

matrix at this narrow-scope retrieval task (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§ 3).

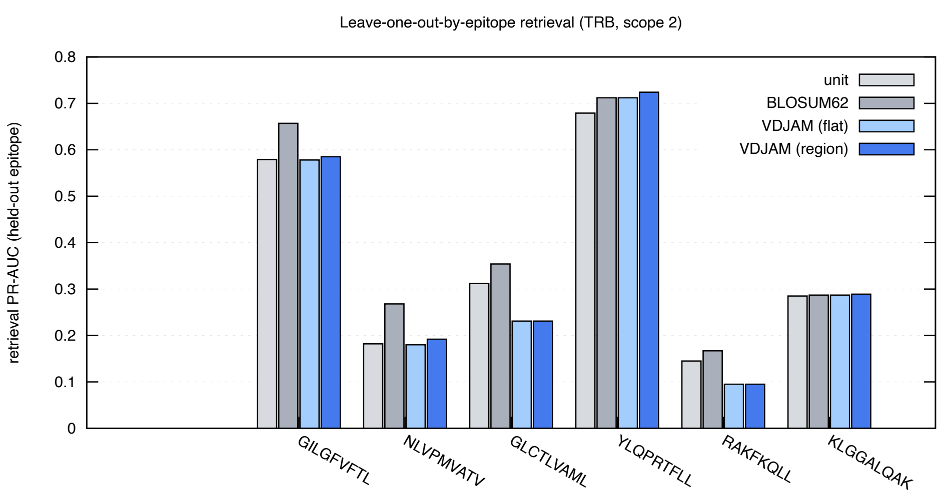

Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. Edit distance, position, and the matrix comparison

Four analyses (`bench/scoring_analysis.py`, `bench/loo_vdjam.py`) settle what scoring actually buys on VDJdb. All figures and numbers below are the *human TRB* example case (the largest, best-populated slice of VDJdb); the same pipeline applies unchanged to human/mouse and to TRA/TRB — point `$VDJDB_SAMPLE`/`$VDJDB_SPECIES` at another export to regenerate.

Hamming distance 1 is the signal:noise optimum.. The fraction of CDR3 neighbours that share an epitope — the precision of a similarity call — falls steeply with edit distance: 0.53, 0.27, 0.19, 0.15, 0.13 at distance $1 \ldots 5$ (Fig. 5). A distance-1 neighbour is twice as likely to be a true co-specific as a distance-2 one and four times a distance-5 one. This reproduces the original VDJdb observation [6] that single-mismatch neighbours are the sweet spot, and it frames everything below: the search ball, not the substitution matrix, is the dominant lever; the matrix only re-ranks within a ball whose purity is already set by its radius.

Central substitutions carry the specificity signal.. Conditioning on *where* a single substitution falls (Fig. 6), the probability that the two CDR3s still share an epitope dips to ≈ 0.41 in the NDN core but rises to ≈ 0.70 toward the J border: a central mismatch most often *changes* specificity (it is contact-proximal and informative), while a border mismatch is usually germline noise (the conserved V/J stems). This is the empirical basis for weighting the centre more — the same direction as the germline-retention weight $1 - w(p)$ (§ 2) — and, encoded directly as a positional weight on BLOSUM62 (the PSSM of § 5), it is the one change that consistently improves retrieval over flat BLOSUM62 (Table 1). It also matches the contacting-region picture of Shcherbinin et al. [5].

The NDN core is glycine-rich.. Central CDR3 positions reach $\sim$ 30–36% glycine in both chains (Fig. 7) — the N-region/D-gene insertion signature — a strongly non-BLOSUM composition. Notably TRA is comparably central-G-rich to TRB, so this is an insertion-diversity feature, not a TRB-D-gene-exclusive one; either way the NDN alphabet is distinctive enough that a general matrix is a poor fit there in principle.

No *standard* matrix beats BLOSUM62.. Re-scoring the held-out-epitope candidates (leave-one-out, § above) with each substitution matrix gives the same verdict at every scope (Table 1, Fig. 4): BLOSUM62 and PAM250 are the best and essentially tied, the structural matrix sits between them and plain Hamming, Hamming (unit) is a strong baseline, and the data-derived NDN-VDJAM — flat or region-weighted — trails them all (region beats flat by a small, consistent, scope-growing margin). So among off-the-shelf and data-derived *amino-acid* matrices, the honest answer to "can we beat BLOSUM62" is no: the substitution matrix is a second-order lever once the ball radius is fixed, and the first-order statistic remains the control-calibrated neighbour count at distance 1 (§ 3).

But position-weighting BLOSUM62 does (the significance PSSM).. Experiment 2 says *where* a substitution falls matters more than *which* substitution it is. We encode that directly as a positional factor on top of BLOSUM62 rather than as a new alphabet matrix. From the position-significance curve (Fig. 6) we take a per-relative-position weight $\omega(p) = (1 - \Pr(\text{same} \mid \text{sub at } p))$, normalised to mean 1 (centre ≈ 1.4, J border ≈ 0.7), and interpolate it to any CDR3 length. `seqtree` consumes this natively: `regions.significance_pssm()` builds a fixed-width `PositionalMatrix.from_weights(BLOSUM62,`

$\lfloor 100\,\omega(p) \rceil$), so the engine scores $\mathrm{pen}(p,a,b) = \omega(p) \cdot \mathrm{BLOSUM62}(a,b)$ — a central mismatch costs $\sim 1.4\times$ its border-position cost. This is the only scoring change that beats flat BLOSUM62, and it does so *at every held-out epitope*: $0.333 \to 0.356$ at distance ≤ 2 and $0.218 \to 0.236$ at distance ≤ 4, winning 8/8 epitopes at both radii (Table 1). The gain is modest in absolute PR-AUC but uniformly signed, exactly because the matrix re-ranks within a ball whose purity is already set by the radius. The remaining lever, the *V-gene* dimension, is taken up in § 6.

A purpose-built TCR matrix (tcrBLOSUM) does not transfer.. Postovskaya et al. [3] report that tcrBLOSUM — a BLOSUM-style matrix counted from same-epitope, equal-length CDR3 pairs — improves epitope-specific TCR clustering over BLOSUM62. Two features make that gain suspect: the counts apply *no* redundancy clustering (so convergent public clonotypes dominate the statistics), and the matrix is validated on the same same-epitope relation it is fit to. Run through our leave-one-out-by-epitope retrieval (their published $\mathrm{tcrBLOSUM_b}$, `bench/tcrblosum_refute.py`), it does *not* beat BLOSUM62: it is slightly worse at both radii (0.303 vs 0.330 at distance ≤ 2, winning 0/8 epitopes; 0.197 vs 0.211 at distance ≤ 4), while their bundled BLOSUM62 reproduces ours exactly (a sanity check on the loader). Reweighting *either* matrix by the central-significance PSSM helps, and it helps BLOSUM62 more than tcrBLOSUM (0.354 vs 0.336). The lesson is consistent: a re-counted amino-acid alphabet does not transfer across epitopes; position does.

mean retrieval PR-AUC	unit	BLOSUM62	PAM250	structural	VDJAM	VDJAM-region	BLOSUM+possig
edit distance ≤ 2	0.306	0.333	0.329	0.314	0.284	0.287	**0.356**
edit distance ≤ 4	0.203	0.218	0.218	0.192	0.177	0.181	**0.236**

Table 1: Substitution matrices compared by leave-one-out-by-epitope VDJdb-vs-VDJdb retrieval (TRB, 8 largest epitopes, micro PR-AUC). Among standard/data-derived amino-acid matrices BLOSUM62 $\approx$ PAM250 lead and VDJAM trails; the only scoring that beats BLOSUM62 is BLOSUM62 reweighted by the positional-significance PSSM (centre > V/J borders), which wins all 8/8 held-out epitopes at both radii.

6. The V-gene dimension

The remaining lever is the V gene. Two questions, both tested by leave-one-out retrieval on human TRB (`bench/vgene_strat.py`, edit distance ≤ 4): is a V match itself a specificity signal, and can it be *softened* to germline-similar V genes?

A V match is a strong, near-binary specificity prior.. Splitting every scored neighbour pair by whether query and hit use the same V family, same-V neighbours share the held-out epitope 53% of the time versus 12% for cross-V — a $\sim$4.3× prior, far larger than any substitution-matrix effect (Table 2). This is why the legacy tool used V as a hard filter, and it argues for carrying V as a strong prior (a soft $S(V)$ term or a per-V null) in a joint V+CDR3 score rather than discarding it.

Position weighting helps both strata equally.. The central-significance PSSM and the germline-retention weighting each add $\approx +0.02$ PR-AUC over flat BLOSUM62 in *both* strata (Table 2). We had expected the gain to concentrate in cross-V pairs, where germline V-flank residues differ and

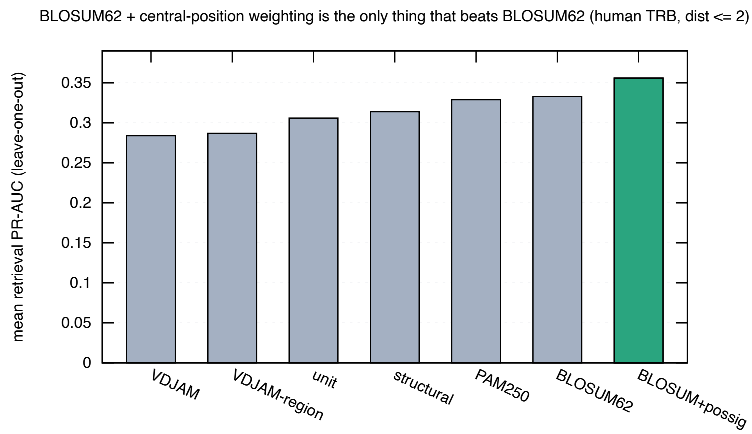

Figure 4: Mean leave-one-out retrieval PR-AUC by scoring (TRB, distance ≤ 2). No standard amino-acid matrix beats BLOSUM62; reweighting BLOSUM62 by the central-position significance factor (green) does.

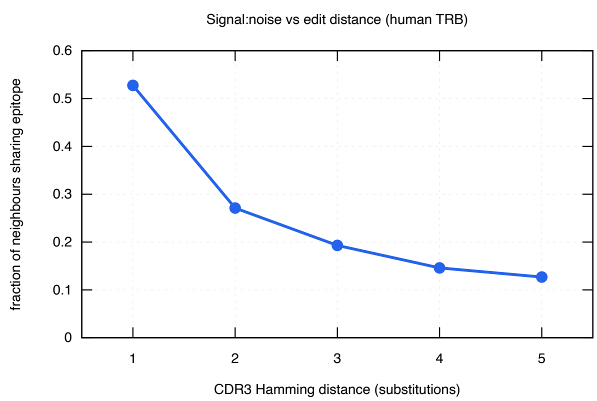

Figure 5: Fraction of CDR3 neighbours sharing an epitope vs Hamming distance (human TRB): distance 1 is the signal:noise optimum [6].

should be down-weighted as noise; instead it is uniform. The weighting is robust rather than cross-V-specific (in relative terms the cross-V gain is larger only because its baseline is far lower).

Germline-loop V-similarity does not recover the prior.. If the same-V advantage were the chemistry of the V-encoded contacts, it should degrade gracefully with germline CDR1+CDR2 similarity, licensing fuzzy V matching (e.g. TRBV5-1≈TRBV5-8). It does not: cross-V co-specificity stays near 12–13% across the whole similarity range, with no monotone gradient and the most-similar bin no higher (Table 3). The V signal is therefore *gene identity*, not contacting-loop similarity — so soft V-clustering by CDR1/CDR2 does not recover it, and a hard or near-hard V constraint remains the right default. (The DE-loop / "CDR2.5" inside FWR3 is not isolated in our germline table; folding it in is a possible refinement, but it is unlikely to bend a flat signal.) The clustering machinery (`match/vgene.py`: germline-loop similarity and single-linkage V groups, 66 → 47 TRB

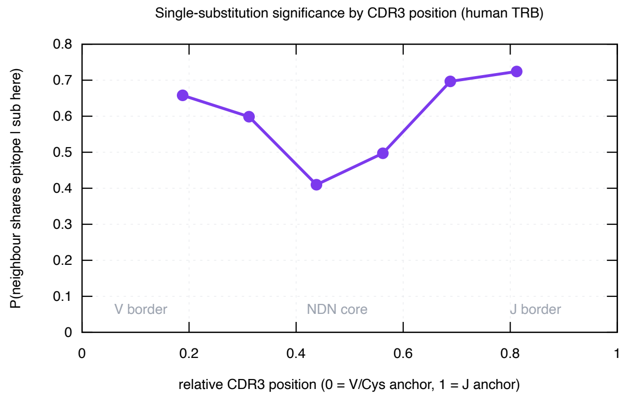

Figure 6: Pr(neighbour shares epitope | single substitution at relative position), human TRB. The NDN core is where a mismatch most changes specificity; the V/J borders are germline noise.

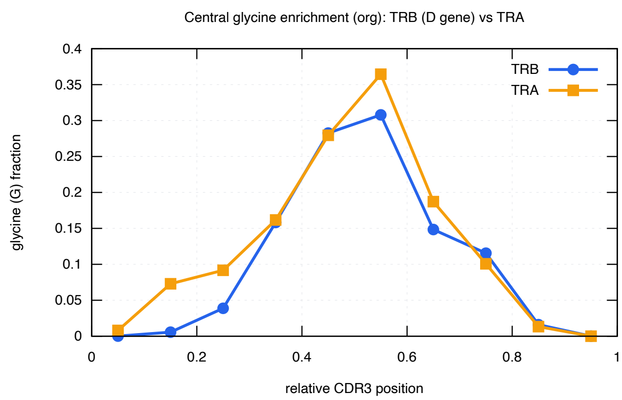

Figure 7: Glycine fraction by relative CDR3 position (human): the NDN core is $\sim$ 30–36% glycine in both TRA and TRB (insertion/D-gene signature).

groups at a 0.8 cut) is shipped for downstream use regardless.

The pattern holds across species, chain and MHC class.. Sweeping every {human,mouse}$\times${TRA,TRB}$\times${M I,MHC-II} cell of VDJdb (`bench/vgene_scan.py`, Fig. 8, Table 4) reproduces both findings everywhere: a V match is a large co-specificity prior (same/cross ratio 1.4–17$\times$), and germline CDR1+CDR2 similarity barely predicts cross-V co-specificity (point-biserial r between -0.06 and $+0.13$ in every adequately powered cell; the single $r = 0.33$ is mouse TRB MHC-II with only 35 cross-V pairs). The near-empty high-similarity bins are themselves telling: germline-similar but non-identical V genes are rare, so CDR1/CDR2 clustering has almost no pairs to act on. We conclude — comprehensively, not just on human TRB — that focusing on CDR1/CDR2 to soften the V match is not supported; V should enter the model as a strong, near-binary prior.

V stratum	pairs	co-specific	flat	+possig	+region
same V	130 273	53.1%	0.671	0.694	0.695
cross V	901 819	12.3%	0.164	0.185	0.182

Table 2: V-gene-stratified leave-one-out retrieval (human TRB, edit distance ≤ 4, pooled micro PR-AUC). A V match is a $\sim$4.3$\times$ co-specificity prior; position weighting (+possig, +region) adds $\approx +0.02$ in both strata.

germline CDR1+CDR2 similarity	cross-V pairs	co-specific
$[0.0, 0.3)$	464 817	11.6%
$[0.3, 0.5)$	352 184	13.0%
$[0.5, 0.7)$	66 838	12.8%
$[0.7, 0.9)$	15 831	16.6%
$[0.9, 1.0)$	2 149	8.3%

Table 3: Cross-V co-specificity does not rise with germline contacting-loop similarity (human TRB): the same-V prior (53%) is not interpolated by V-gene similarity, so soft V-clustering does not recover it. The V signal is gene identity, not loop chemistry.

7. Worked examples

Pairwise (different lengths).. `seqtree` aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved `CASS` V-anchor and `DTQYF` J-motif with substitutions and indels confined to the central NDN core:

```
query (ref): CASSPRSGETQYF   epitope GILGFVFTL   (unit / edit-distance alignment)

vs CASSNRSGETQYF  (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
  CASSPRSGETQYF
  |||| ||||||||
  CASSNRSGETQYF

vs CASSIRSGESTQYF  (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)
  CASSPRSGE-TQYF
  |||| |||| ||||
  CASSIRSGESTQYF

vs CASSLFSETQYF  (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
  CASSPRSGETQYF
  ||||  | |||||
  CASSLFS-ETQYF

vs CASSLGETQYF  (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
  CASSPRSGETQYF
  ||||   ||||||
  CASS--LGETQYF
```

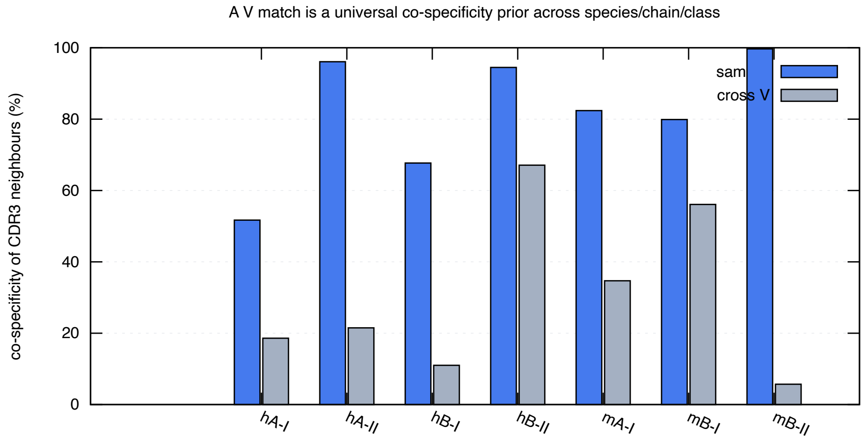

Figure 8: Same-V vs cross-V CDR3-neighbour co-specificity across species/chain/MHC-class cells of VDJdb (h/m = human/mouse; A/B = TRA/TRB; I/II = MHC class). A V match is a universal co-specificity prior; cross-V neighbours are far less likely to share the epitope, and germline CDR1/CDR2 similarity does not bridge the gap (Table 4).

cell (species / chain / class)	same-V	cross-V	ratio	r(vsim, co-spec)
human TRA MHC-I	53.8%	18.4%	2.9	0.086
human TRA MHC-II	96.1%	10.2%	9.4	0.126
human TRB MHC-I	69.0%	9.7%	7.1	0.046
human TRB MHC-II	94.2%	67.2%	1.4	0.039
mouse TRA MHC-I	82.8%	34.8%	2.4	0.047
mouse TRB MHC-I	80.6%	56.3%	1.4	−0.056
mouse TRB MHC-II	99.7%	5.7%	17.4	0.331

Table 4: Per-cell same-V vs cross-V co-specificity and the point-biserial correlation between germline CDR1+CDR2 similarity and co-specificity among cross-V pairs (unit-cost candidates, top 6 epitopes/cell, $\geq$ 30 CDR3/epitope; mouse TRA MHC-II omitted, n too small). The V prior holds everywhere; $r \approx 0$ wherever it is well powered.

```
vs CASSPWSGSQETQYF  (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
  CASSPRSG--ETQYF
  ||||| ||  |||||
  CASSPWSGSQETQYF
```

Multiple (same length).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 9–10): columns are shaded by conservation, dark at the germline-determined `CASS` prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [5] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

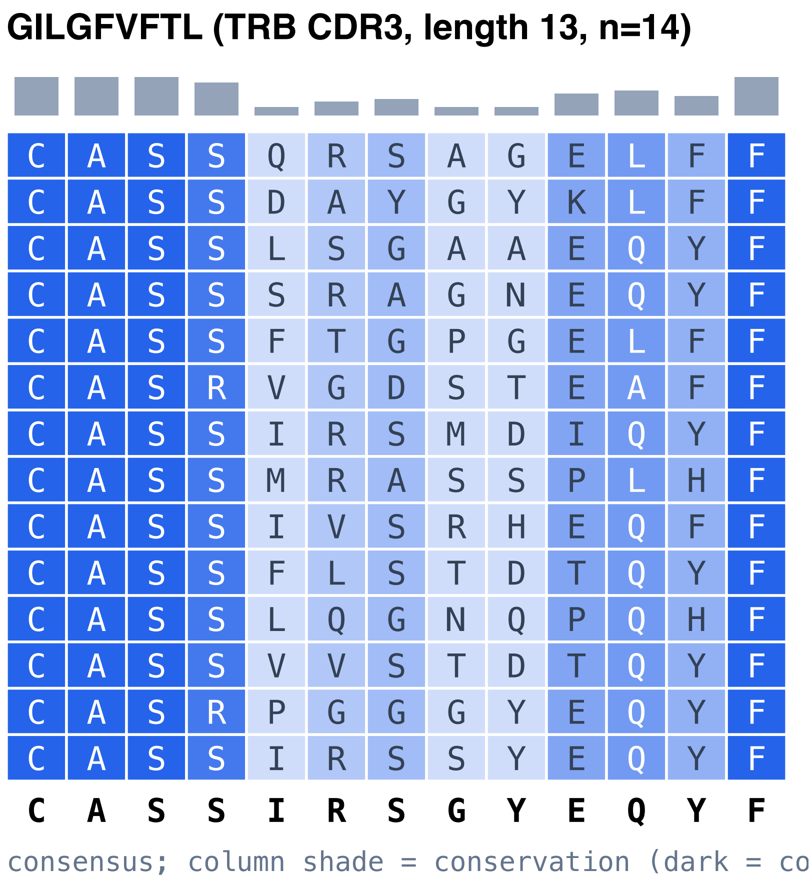

Figure 9: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

8. Limitations and roadmap

The data-derived NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and does not beat BLOSUM62 at any scope; region-aware germline+trimming weighting gives a small, consistent gain over the flat matrix but not over BLOSUM62. The substitution *alphabet* is thus a second-order lever. What does move the needle is *position*: reweighting BLOSUM62 by the central-significance PSSM beats flat BLOSUM62 at every held-out epitope (§ 5), confirming that for CDR3 *where* a mismatch falls matters more than *which* residue it is. The *V-gene* dimension (§ 6) turns out to be a strong but near-binary lever: a V match is a $\sim$4.3$\times$ co-specificity prior, yet it does *not* interpolate with germline contacting-loop similarity, so soft V-clustering by CDR1/CDR2 does not recover it and a near-hard V constraint stays the default. The open work is therefore to model V as a strong prior in a joint V+CDR3 E-value rather than to soften the V match, plus a per-chain treatment (TRA vs TRB). Separately, the control's V–J usage must match the query repertoire's or the null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes.

A further direction is to predict, *a priori*, which epitopes are intrinsically easy versus hard to annotate — the wide spread of per-epitope PR-AUC in our experiments (e.g. the convergent, "fea-

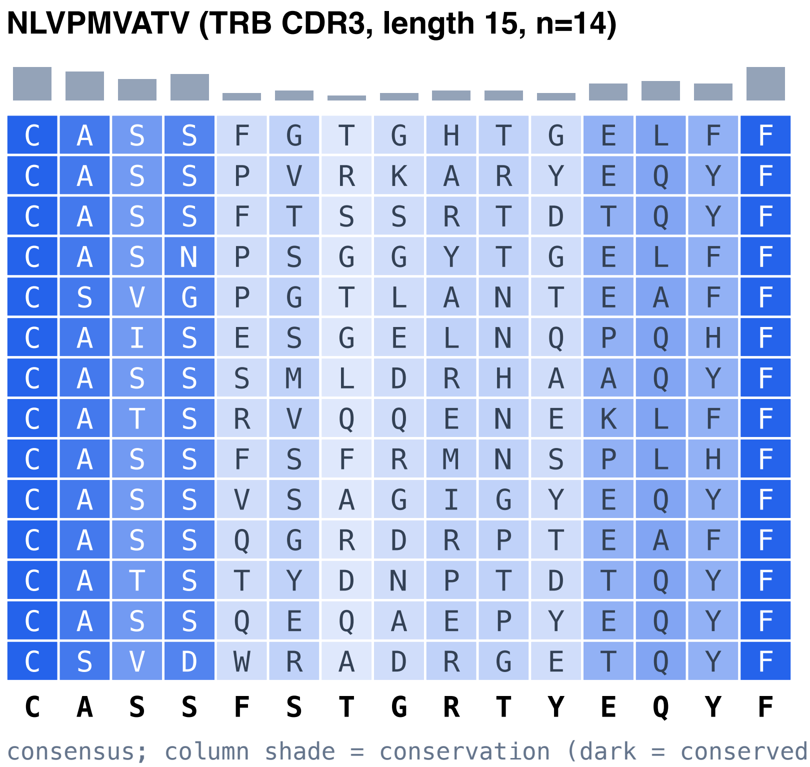

Figure 10: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

tured" CMV NLV vs the diffuse, "featureless" influenza GIL) is not noise but biology. A pMHC with prominent TCR-facing features selects a focused, motif-rich repertoire that clusters cleanly, whereas a featureless one is recognised by structurally diverse TCRs in varied docking modes and yields no tight motif [8, 9, 7, 2]; clustering-based specificity inference succeeds or fails accordingly [1]. A model that scores epitope "annotability" from repertoire convergence (and, where available, pMHC structural features) would let us report calibrated confidence per epitope and triage GIL-like hard cases — a planned addition to the benchmark. These extensions, and the full comparison against existing tools, are ongoing work.

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